

Smart

**And Grid-Interactive
Electric Panels**

Table Of Contents

Introduction

Let us tell you a little about us.

Hardware, Software, and Mechanical

PCB Design

Prototyping and Testing

Administrative Content

Introduction

Meet The Team

Jack Dwyer
EE

Nick Sally
EE

Oscar Rodriguez
CPE

Abraham Alcala
CPE



Problem

lack the capability to interact with the grid

renewable energy sources challenge stability of the grid

Little to no existing products

Motivation

enhances the adaptability and efficiency of residential electrical systems

Allow homeowners to actively participate in energy management

empower consumers to actively participate in the transition to a cleaner energy grid.

Basic Goals

- Design and execute a mobile interface
- Automatic Routines for frequency sensing
- Easy installation

Advanced Goals

- Robust user security & verification for mobile app
- User defined scenes for breaker management
- Pathfinding for efficient switch flipping

Stretch Goals

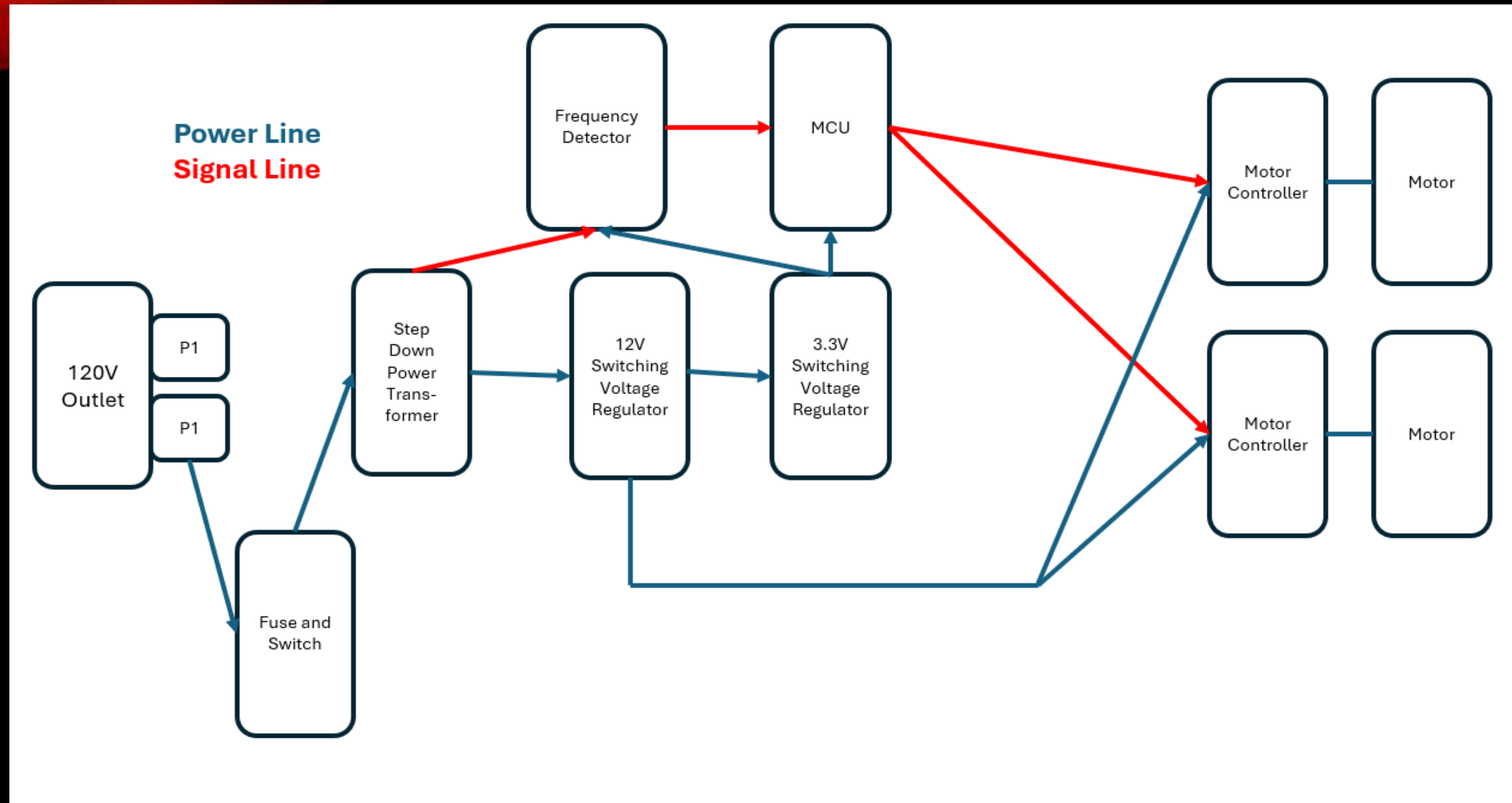
- Current sensing
- Robust on-device MCU communication security

Goals & Objectives

Requirement	Specification	Value
Frequency Detection	The project should be able to detect frequency variation in the electrical grid.	60 ± 1 Hz
Breaker Switching Time	Once frequency is out of bounds or a manual override is present, the circuit breaker is flipped quickly	<10 seconds
Mobile Latency	The time it takes for the product to start flipping switches after the mobile app command will be quick	<2s
Physical Size	Height and length shall not exceed that of the breaker receptacle itself. Depth as measured from the mounting wall	<8"
Product Weight	Minimizing weight is always valuable for consumer products.	<30lbs
Product noise	Product should be quiet. No louder than the switching of the switch itself.	<70dB
Product Power	Minimizing idle power further improves marketability, it is still inconclusive as to how to test this. Product will have a low quiescent power.	<10W

Engineering Requirements

Hardware



Hardware Block Diagram

	166P18	1182N9	241-8-20	187F20
Description	Transformer, low voltage, 115V 60 Hz., 90VA, 18 C.T., 5A	Power transformer, toroid, 80 VA, with 9V @ 8.88A secondary	PWR XFMR LAMINATED 100VA CHAS MT	PWR XFMR LAMINATED 100VA CHAS MT
Price	\$39.46	\$64.00	\$20.91	\$32.18
Secondary Voltage	18VAC CT	18VAC CT	20V	20V
Secondary Current	5A	4.44	5A	5A
Manufacturer	Hammond Manufacturing	Hammond Manufacturing	Signal Transformer	Hammond Manufacturing
Stocked	Yes	Yes	Yes	Yes

- All of these transformer meet out needs
- Settled on 241-8-20 for slightly higher secondary voltage and significantly lower cost

Transformer Selection Matrix

	SD1 Options*				SD2 Final Choice*
	LM5148RGYR	TPS56637RPAR	TPS56837HRP AR	LMR51450SDR RR	LM2679T- 12/NOPB
BOM Count	17	15	14	13	10
Estimate Cost	\$1.93	\$2.75	\$3.17	\$2.67	\$2.53
Efficiency	94.588%	96.398%	96.882%	92.963%	91.393
Switching Frequency	2.075MHz	561.958kHz	480.71kHz	349.674kHz	260kHz
IC Power Dissipated	1.22W	2.24W	1.93W	4.54W	5.65W
In Stock	Yes	Yes	Yes	Yes	Yes

*Info collected
from Webench
circuit

- Chose the LM2679T-12/NOPB
 - Lowest Switching frequency
 - Minimal amount of external parts
 - Known to be a robust and reliable component
 - Exposure in E2 lab

12V PMIC Selection Matrix

SD1 Options*

	TPS561201DD CR	TPS561208DD CR	TPSM861253R DX
BOM Count	8	8	7
Estimate Cost	\$0.71	\$0.71	\$0.69
Efficiency	92.567%	92.567%	83.124%
Switching Frequency	520.109kHz	520.109kHz	1.4 MHz
IC Power Dissipated	0.26W	0.26W	0.67W
In Stock	Yes	Yes	Yes

*Info collected
from Webench
circuit

SD2 Final
Choice

MIC4680- 3.3YM
5
\$2.96
80%
200kHz
0.7W
Yes

- Chose MIC4680-3.3YM
 - Has the lowest switching frequency
 - Weeks recommended we do not use TI regulators

3.3V Selection Matrix

	TLV7011	TS3021	LM2903-Q1	TLV703x-Q1
Description	Analog Comparators Low power, small size comparator with push-pull output 5-SC70 -40 to 125	Analog Comparators Rail-to-rail 1.8 V high-speed comparator	Analog Comparators Dual Comparator Diff	Analog Comparators Automotive nano power comparator with push-pull output 5-SC70 -40 to 125
Output Type	Push-Pull, Open-Drain	Push-Pull	Open-Collector outputs.	Push-Pull, Open-Drain
Propagation Delay	<350ns	<100ns	<2μs	<3 μs
Hysteresis	4.2mV	Unknown	Unknown	6.5mV
Price (\$)	0.65	1.76	0.64	0.86

- Chose the TLV7011
 - Came with a leaded package
 - Micro-Watts of power consumption
 - Existing circuits to model from

Comparator Selection Matrix

	DRV8834	A4988	DRV8825	TMC5160	TB6600
Continuous current (A)	1.5	1.2	1.1	3	3.5
Micro stepping	1/32	1/16	1/32	1/256	1/32
Special features	low input voltage			CoolStep, StallGuard2, SpreadCycle, StealthChop2	Opto-isolators

Motor Controller Selection Matrix

	Pi 4	Pi Pico	ESP32	STM32	Arduino Nano
RAM	1 GB	264 KB	520 KB	512 KB	2 KB
Clock Speed	1.8 GHz	133 MHz	240 MHz	72 MHz	16 MHz
Core Count	4	2	2	2	1
Cost	~\$45	~\$6	~\$11	~\$11	~\$20

- Chose the ESP32
 - Familiarity due to EEE 4775
 - Cheaper yet performative package
 - Good documentation to support PCB design

MCU Selection Matrix

Software

Database Matrix

	PostgreSQL	MongoDB	MySQL
Complexity	Very Low	Low	Moderate
Speed	Moderate	High	Moderate
Documentation	High	High	High



Server Languages

	Golang	Node.js	PHP	Python
Complexity	Moderate	Low	Moderate	Low
Speed	High	High	Moderate	Low
Documentation	Moderate	High	Moderate	Moderate



Backend Comparisons

Framework Matrix

	Flutter	Angular	Vue.JS	React
Complexity	Moderate	Moderate	Low	Low
Speed	High	Moderate	High	High
Documentation	Moderate	Moderate	Moderate	High

Communication Matrix

	WiFi	Bluetooth	LoRaWAN
Data Rate	High	Moderate	Low
Power Expenditure	High	Low	Low
Range	High	Low	Moderate



App Development Comparisons

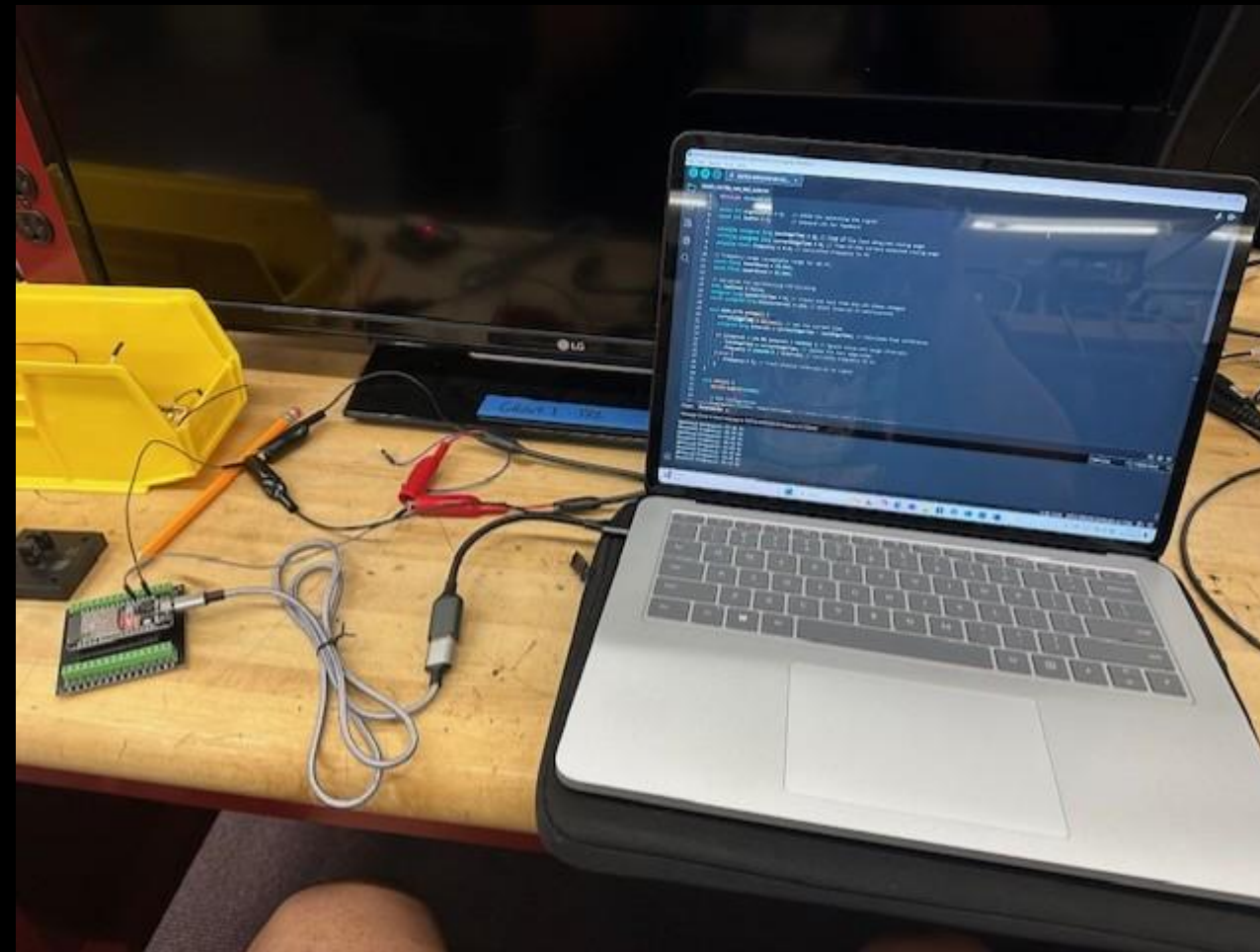
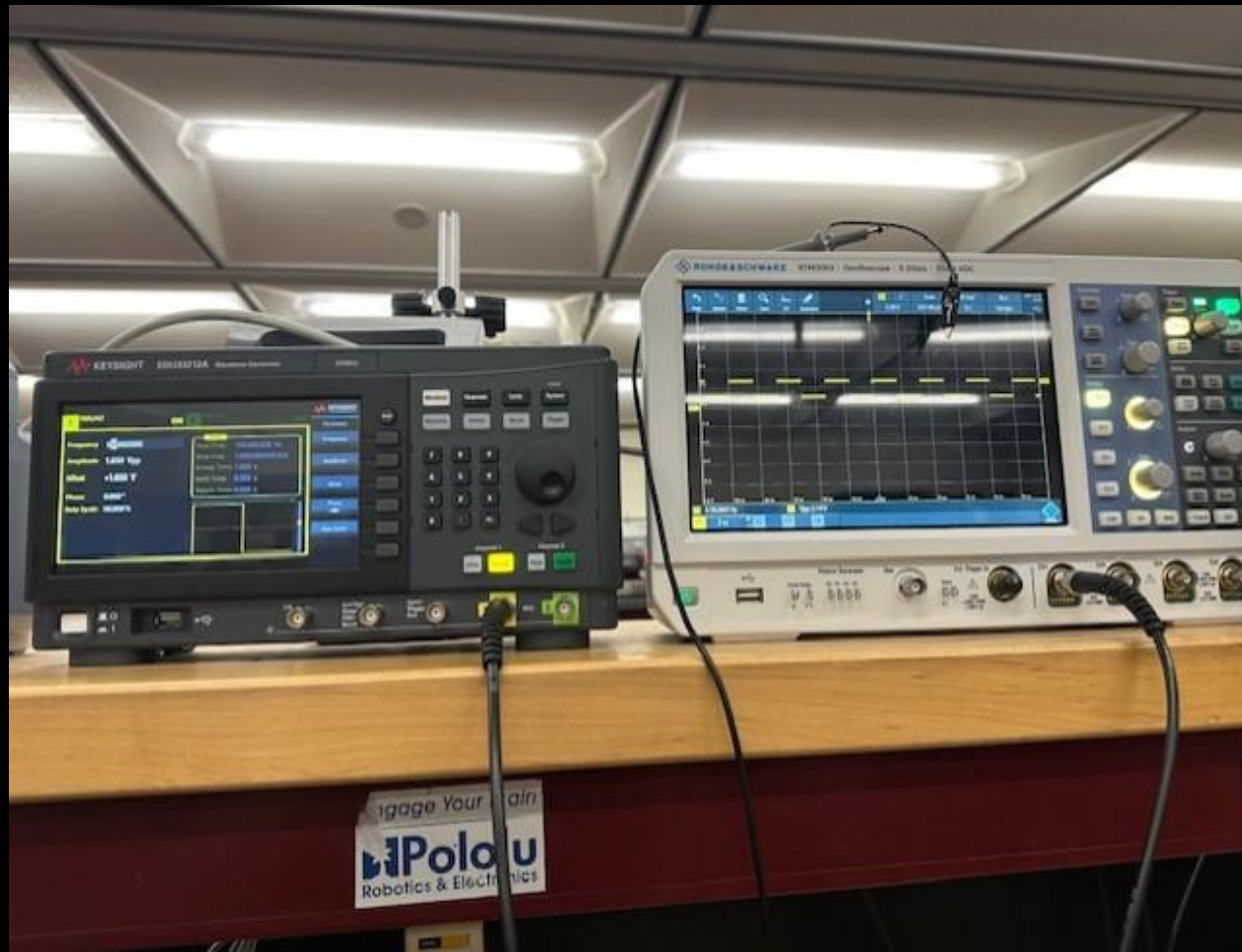
Languages

	C	C++	Python	Rust
Complexity	Moderate	Moderate	Low	High
Speed	High	High	Moderate	Very High
Documentation	Moderate	Moderate	Moderate	High
Libraries	Moderate	High	High	Moderate



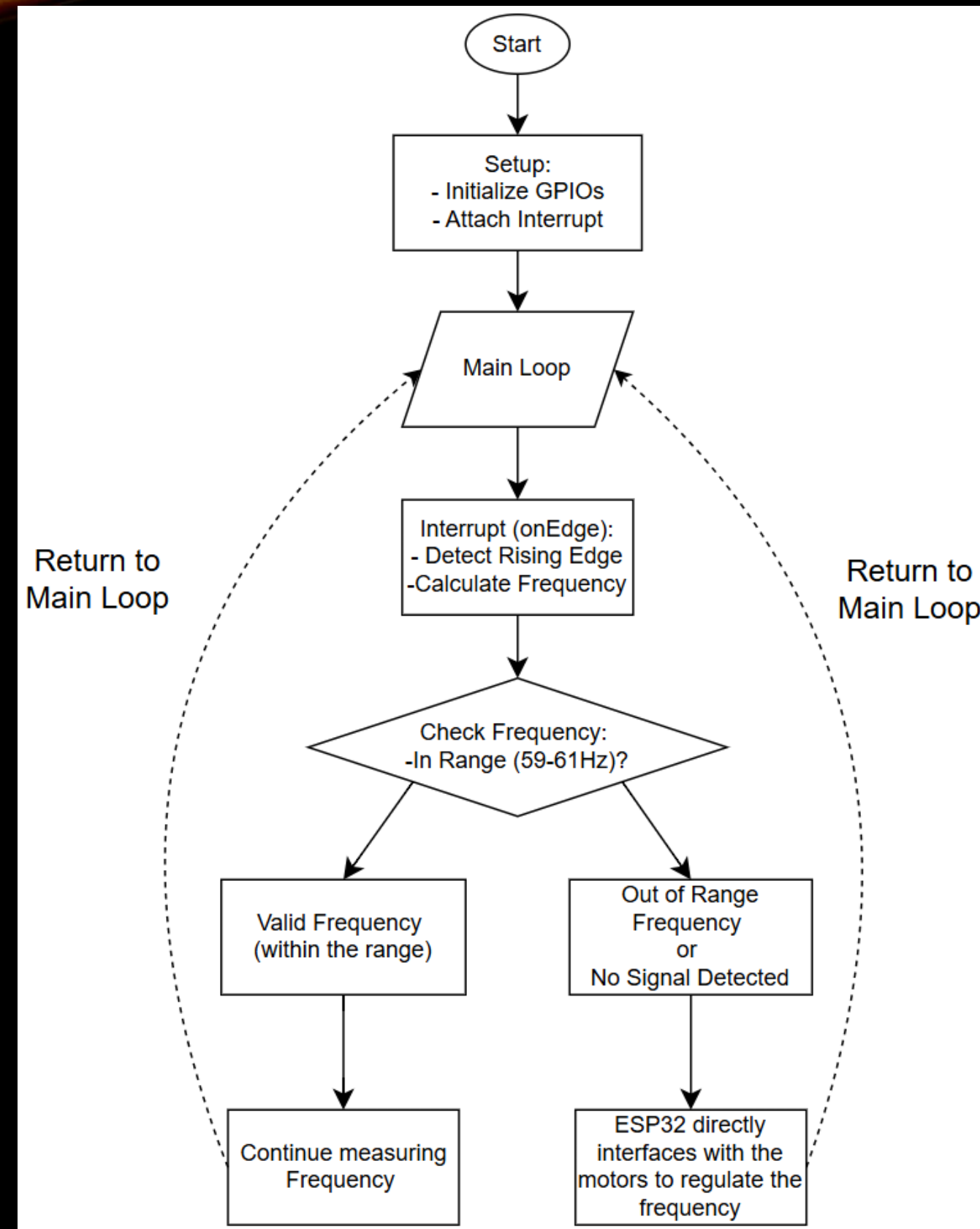
Embedded Software Comparison

Prototype ESP32 Frequency Detection Software

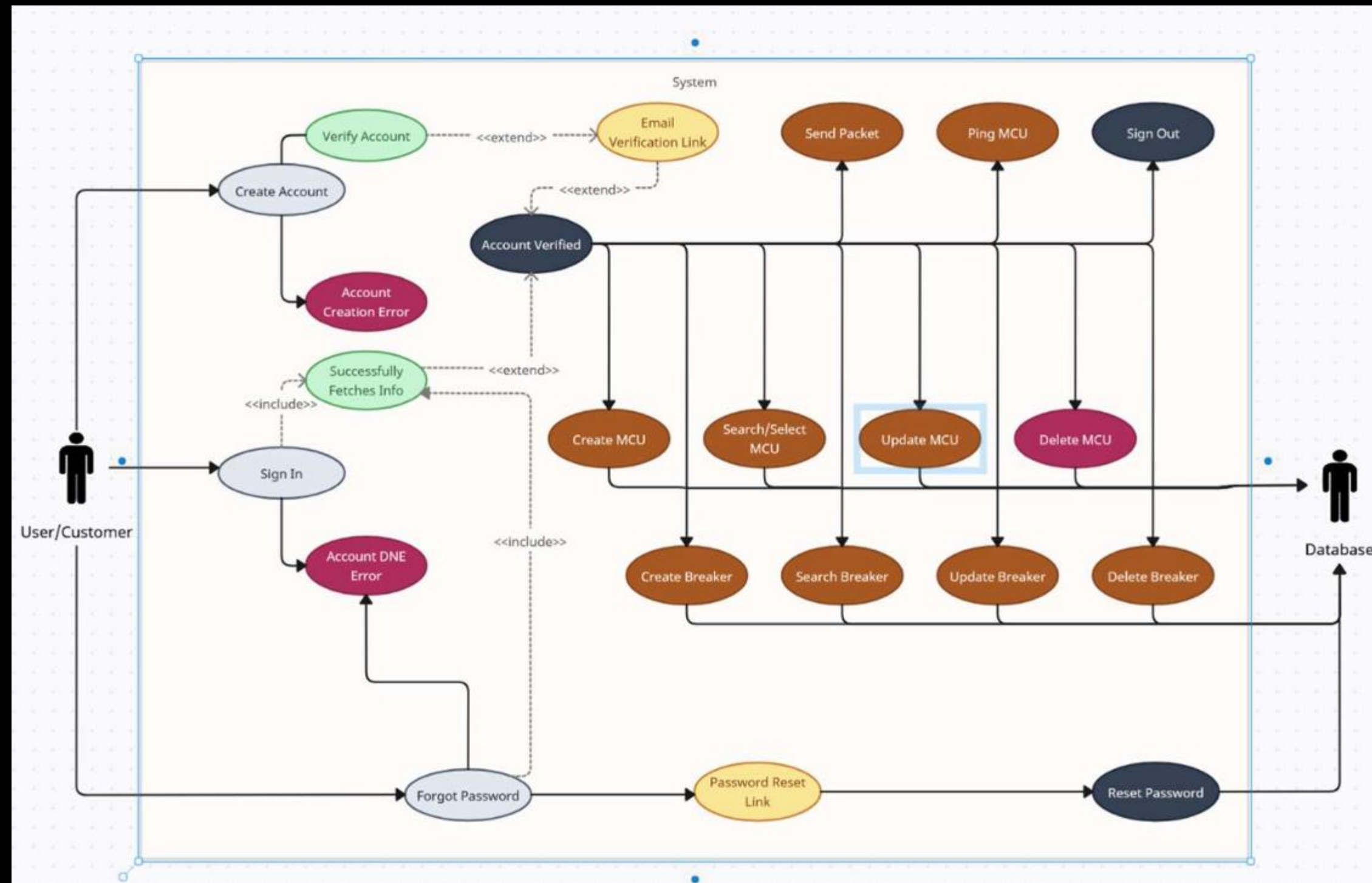


- The Frequency Detection software employs an interrupt-based approach to calculate the frequency of an input signal.
- Rising Edges of an input signal are detected utilizing the ESP32's `attachInterrupt()` function.
- The time interval between consecutive rising edges is measured in microseconds using the ESP32's `micro()` function.
- This interval is then utilized to calculate the frequency using the formula: $f = 1,000,000 / \text{Interval}$

Frequency Detection Software: How it works?



Frequency Detection Software Flowchart



Use Case Diagram

```
1  {
2    "deviceId": "X",
3    "command": "Y",
4    "breaker": "Z"
5  }
```

Input JSON

- X payload denotes a UID that each device upon registration is given
- Y payload denotes a string that tells the ESP32 what it is doing
- Z payload denotes the specific breaker to be toggled

```
Body  ▾  ↻  200 OK • 405 ms • 283 B • 🌐
  {} JSON ▾  ▶ Preview  🔗 Visualize ▾
1  {
2    "deviceId": "i",
3    "resp": "j",
4    "breakerState": "k"
5  }
```

Output JSON

- I payload denotes same as X payload
- J payload denotes a successful response.
- K payload denotes the breakers new state.

Packet Schema

Mechanical



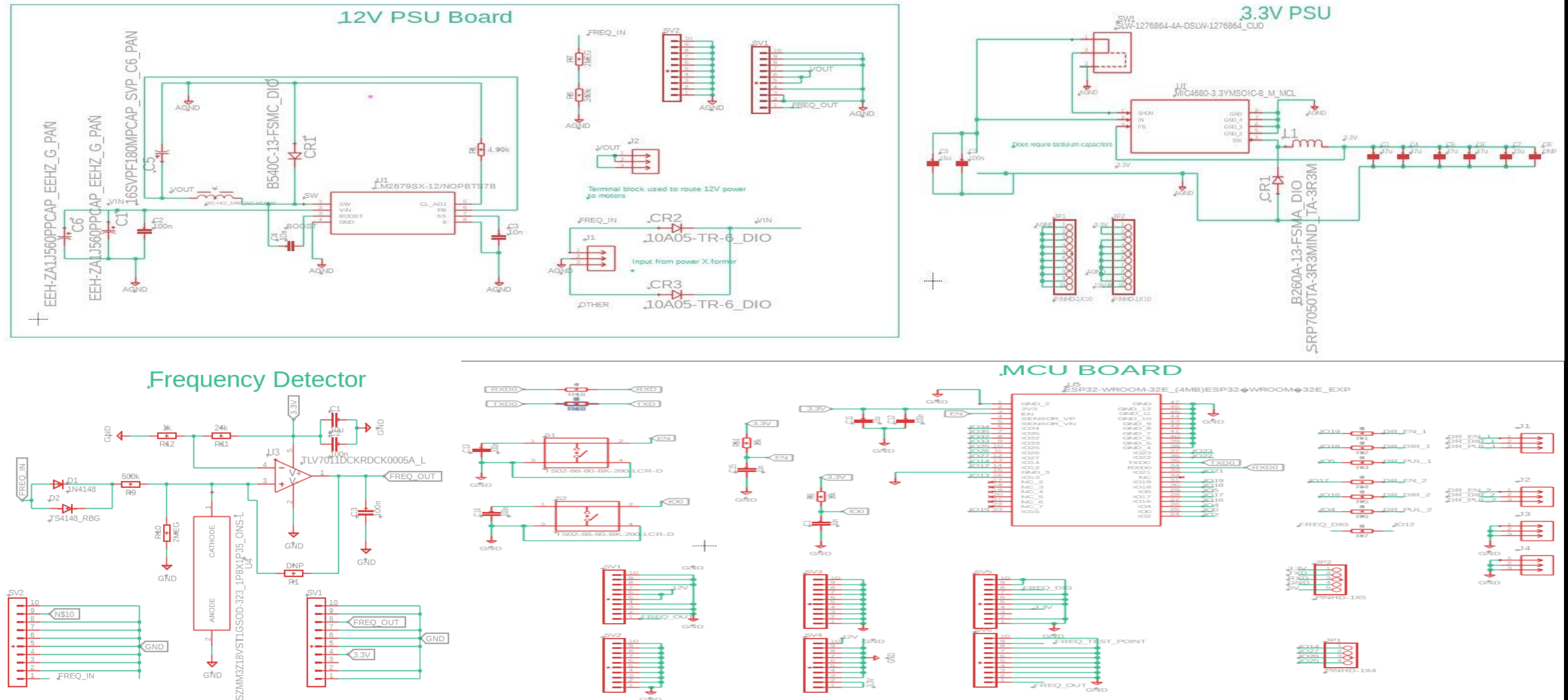
Mechanical System Diagram



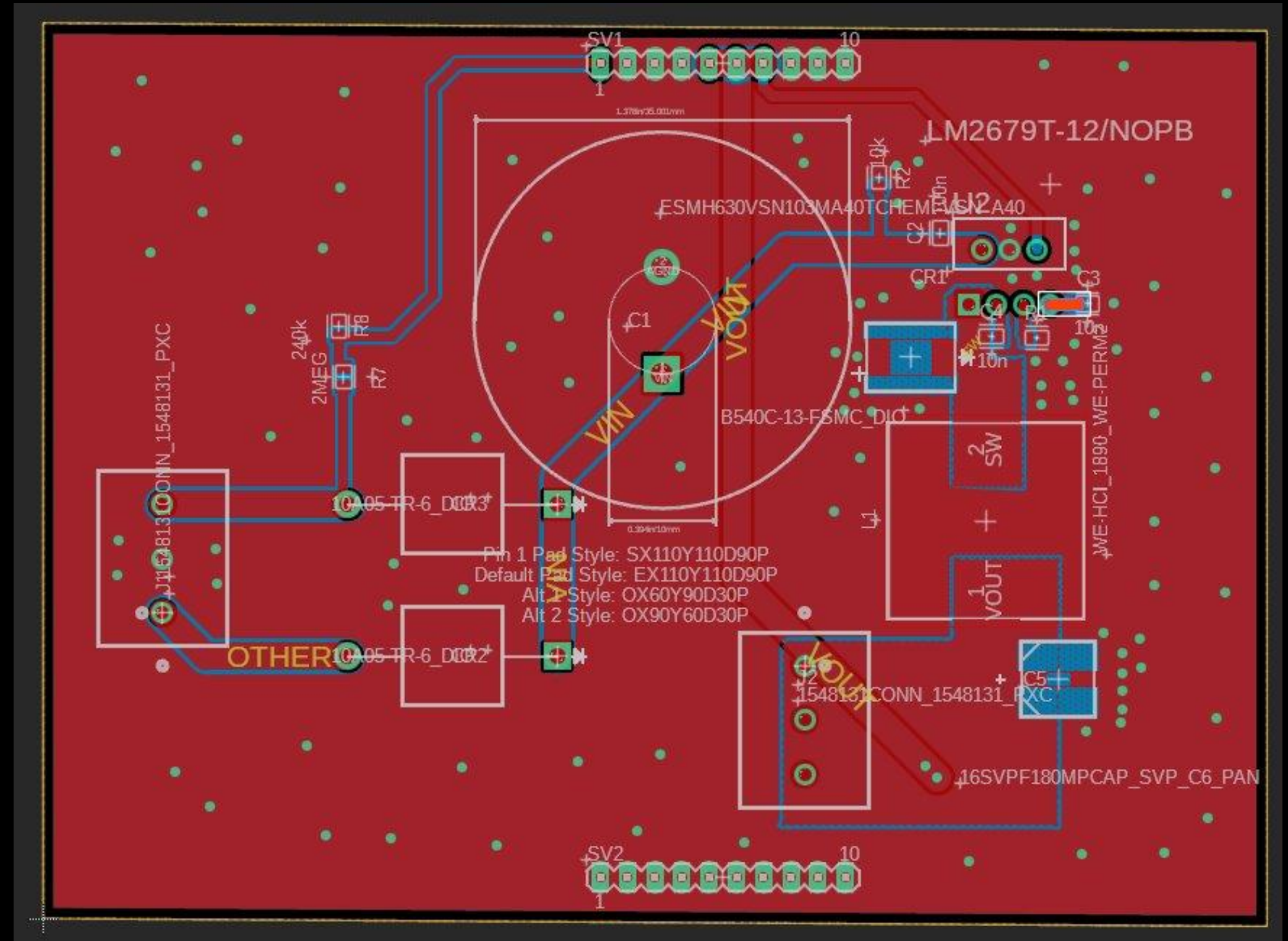
- Gantry based
- Linear rails and lead screws
- Modular design

PCB Design

OVERALL ELECTRICAL SCHEMATIC

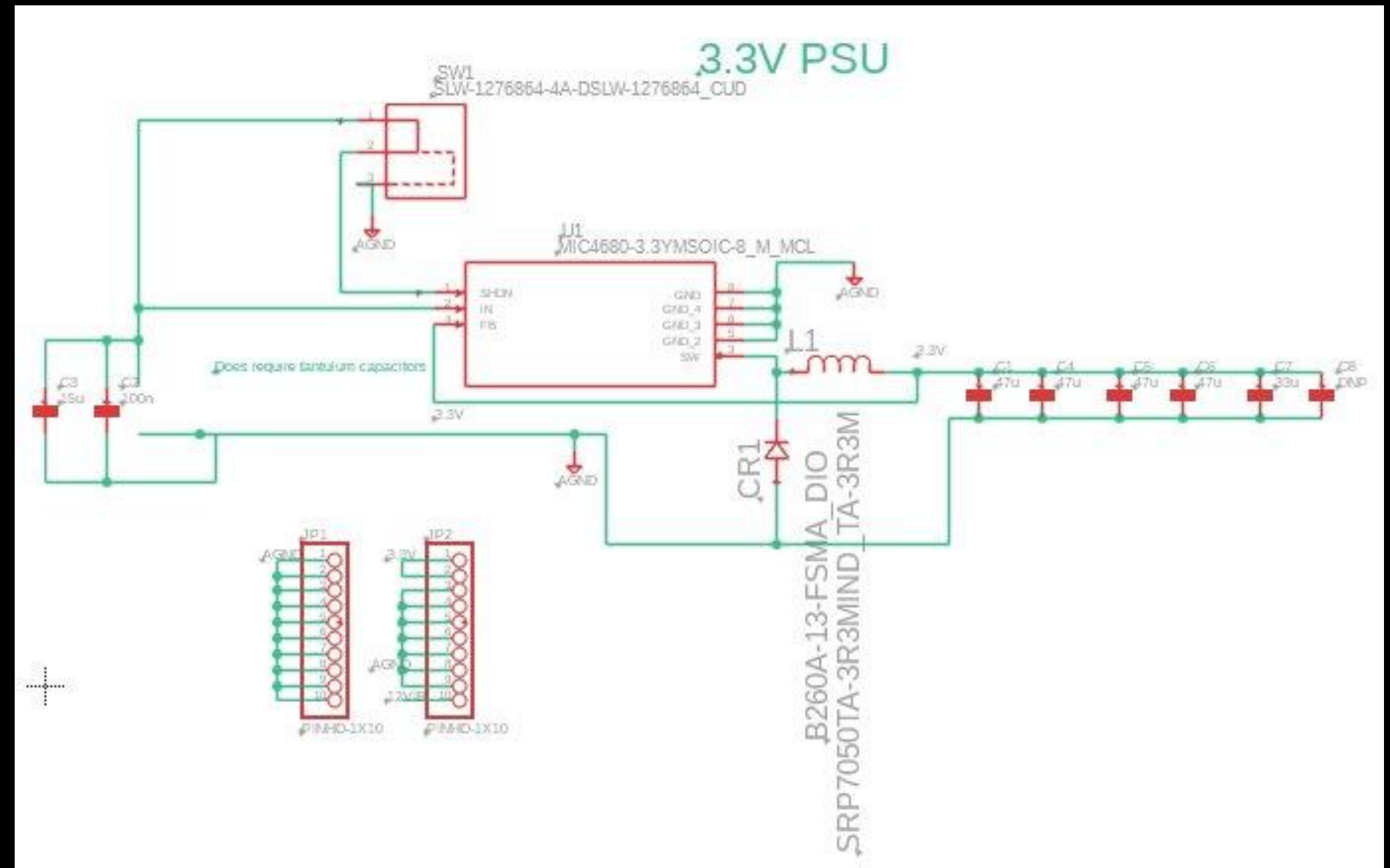


- Design notes:
 - Large DC rectification capacitor
 - Large output poor for high current
 - Motor power fed from terminal block
 - Other power routed through pin headers
 - Input from transformer fed into terminal block



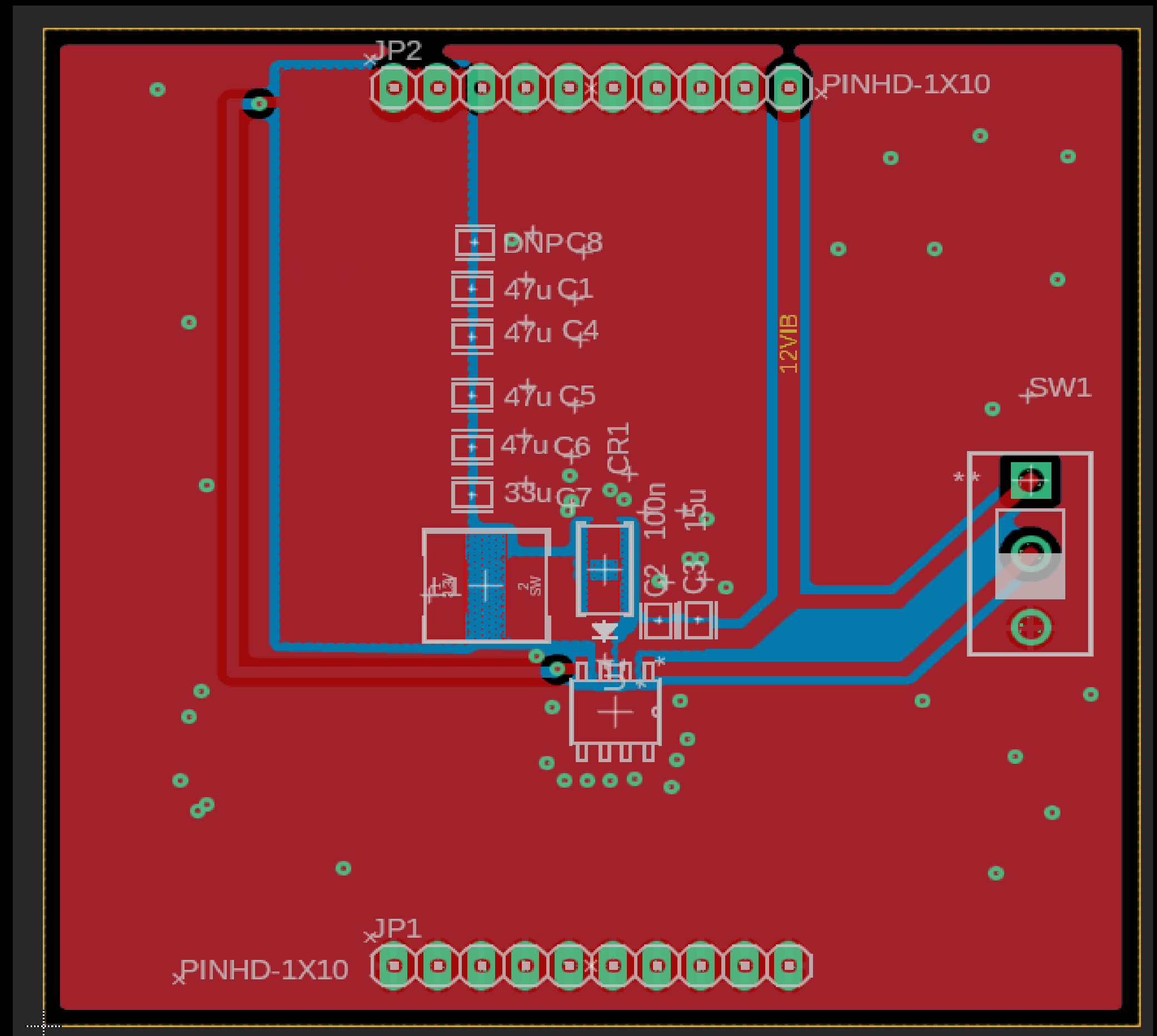
3.3V PSU SCHEMATIC

- Design Note:
 - Original Design from Microchip Technology used tantalum capacitors for low ESR. We changed them for an array of ceramics for the same effect
 - SW3 is an on/off switch



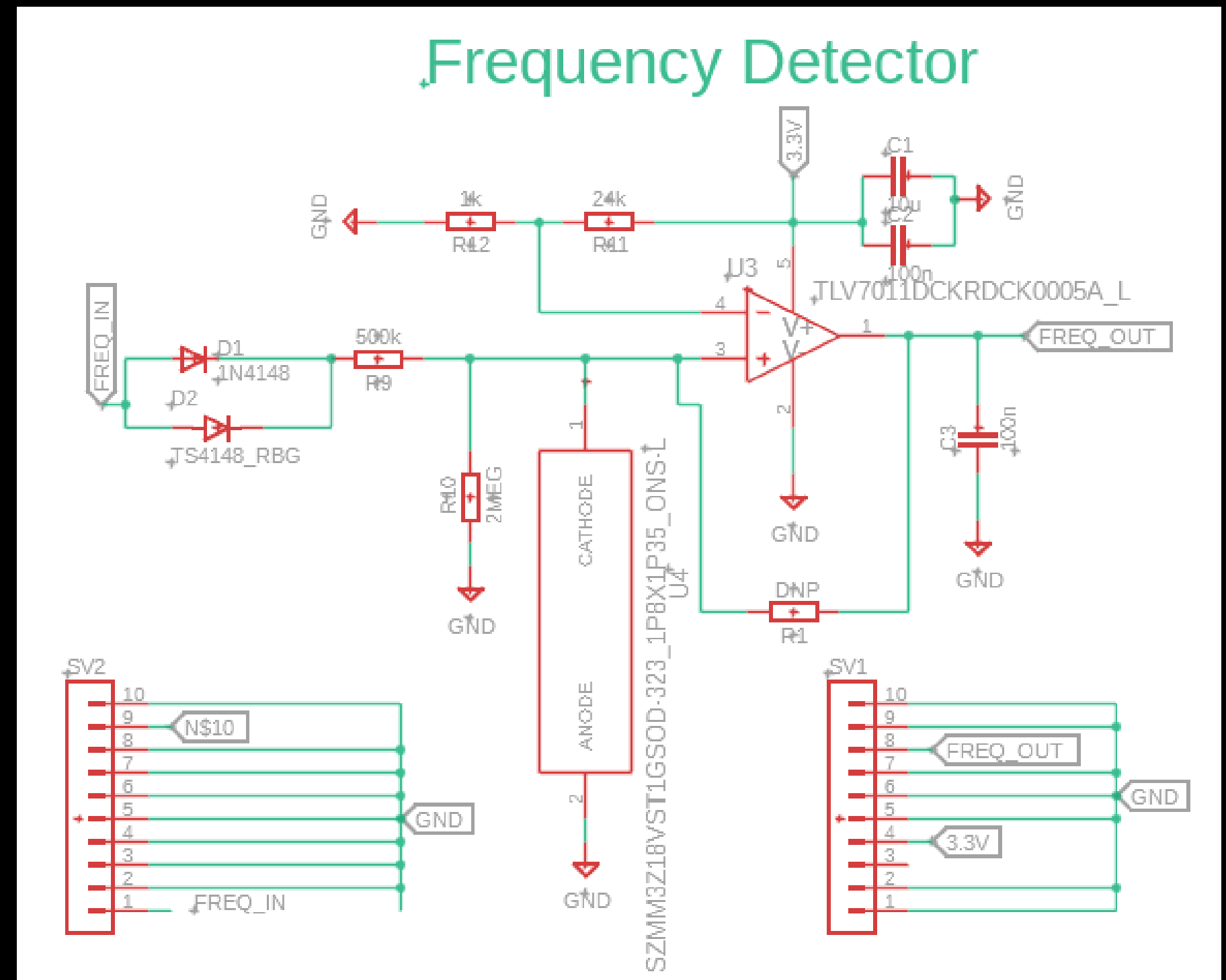
3.3V PSU BOARD

- Design Notes:
 - Large output copper pour for maximum current output
 - Layout follows recommendation from datasheet
 - Slide switch can be bypass if necessary



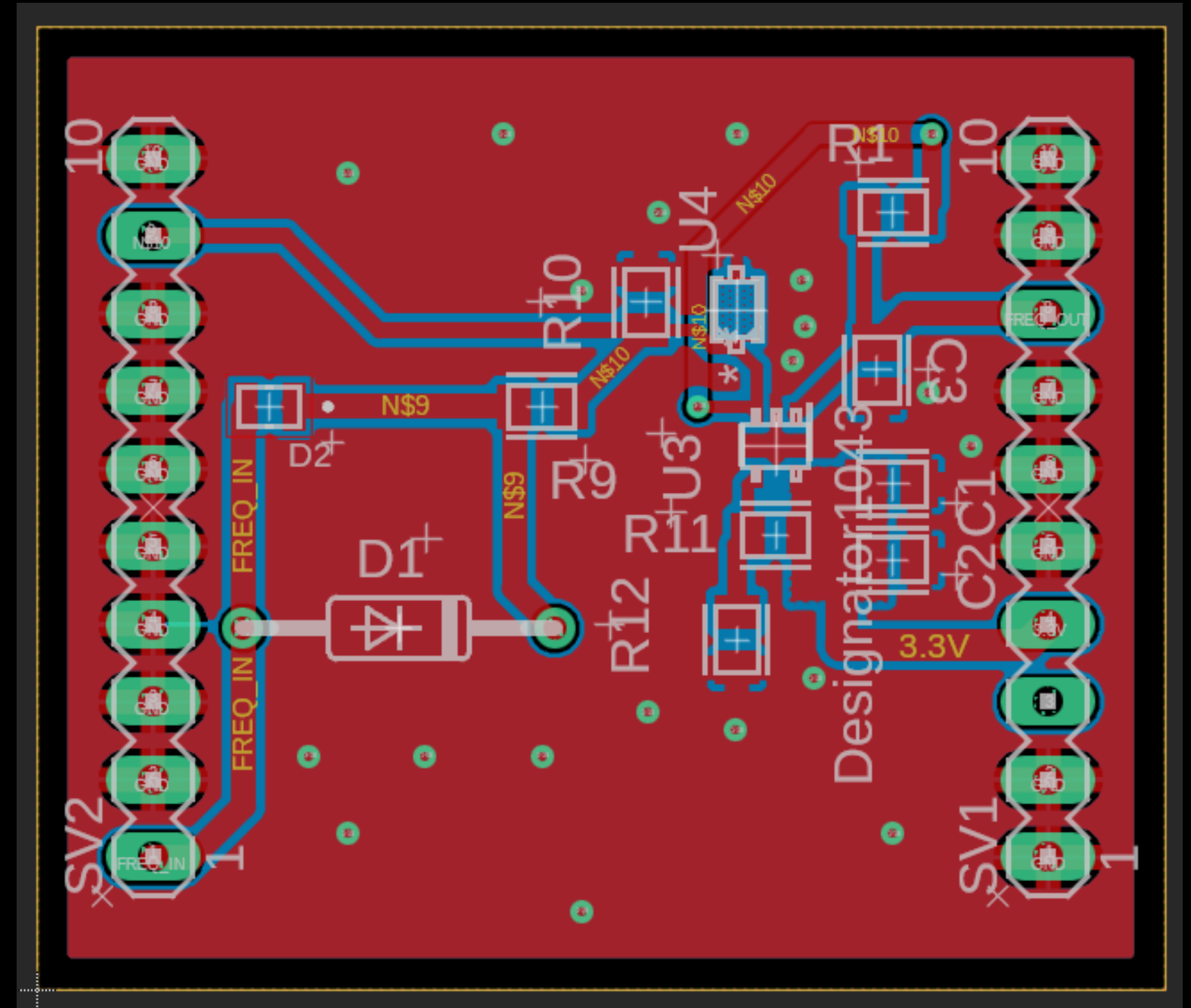
FREQUENCY DETECTOR BOARD

- Design notes
 - Based on TI design for similar application
 - Three diodes listed, but only one populated
 - Additional hysteresis is possible with R1
 - Cross-zero point is at ~100mV



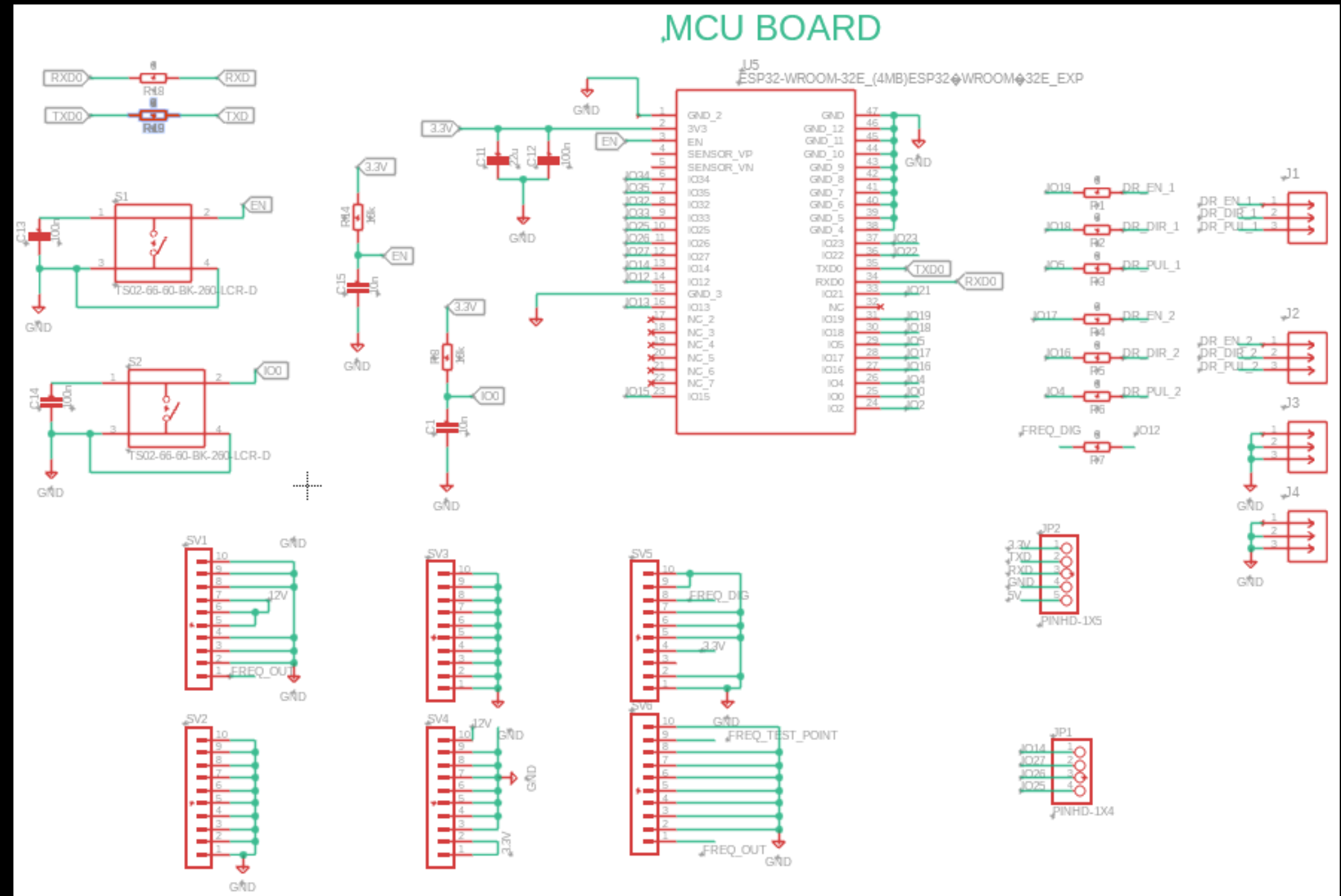
FREQUENCY DETECTOR BOARD

- Different diode package options provide additional flexibility
- Input is fed from high impedance voltage divider from 12V PSU board



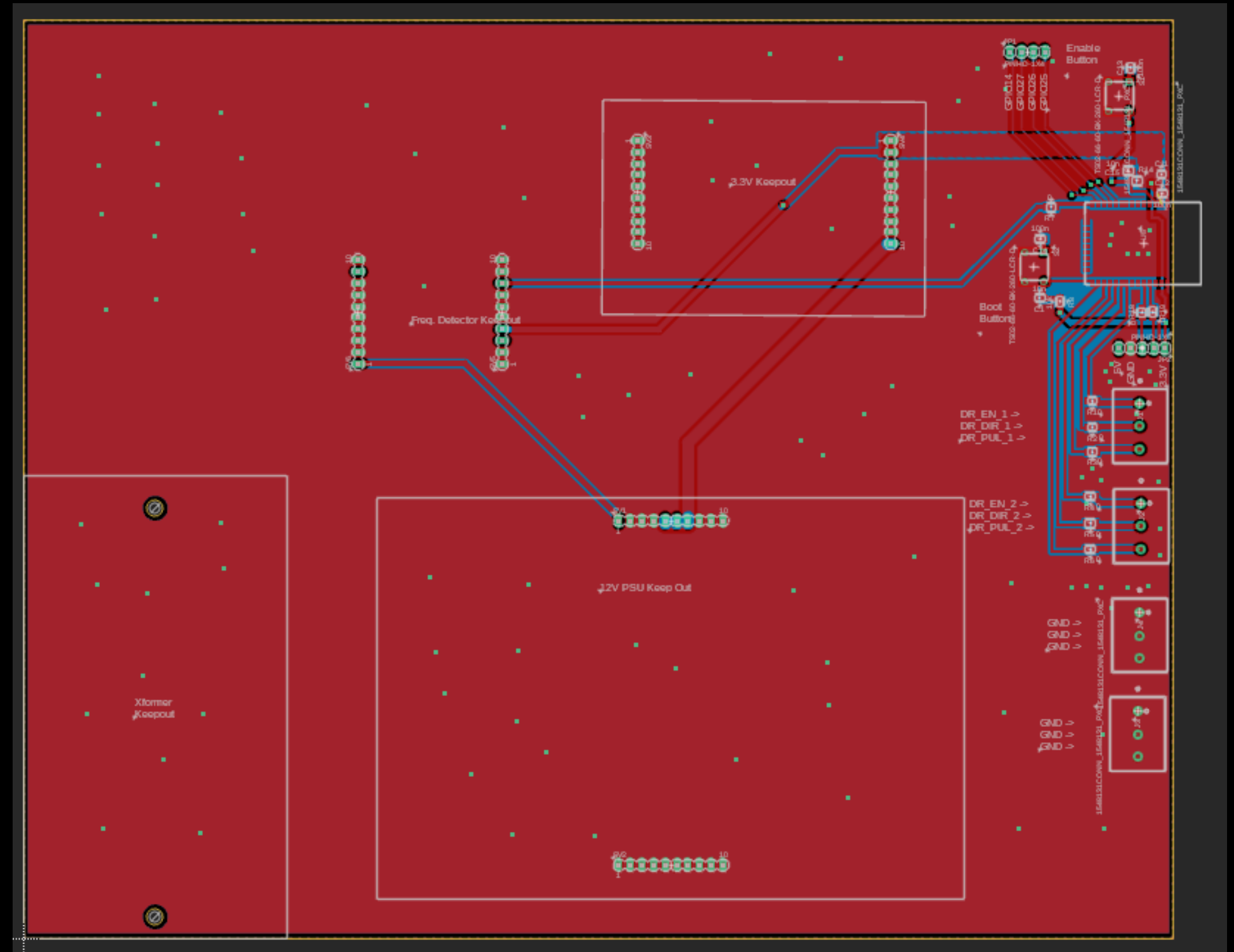
MCU SCHEMATIC

- Where the other three boards will be mounted on
- Programming will be done with HiLetgo CP2102 USB 2.0 to TTL Module Serial Converter Adapter Module
- Terminal blocks feed the command lines to the motor controllers



MCU BOARD

- Terminal blocks have 0-ohm resistors for added flexibility
- Large copper pour for 3.3V input to ESP32
- ESP32 antenna overhands board
- Additional unused space can house motor controllers if necessary

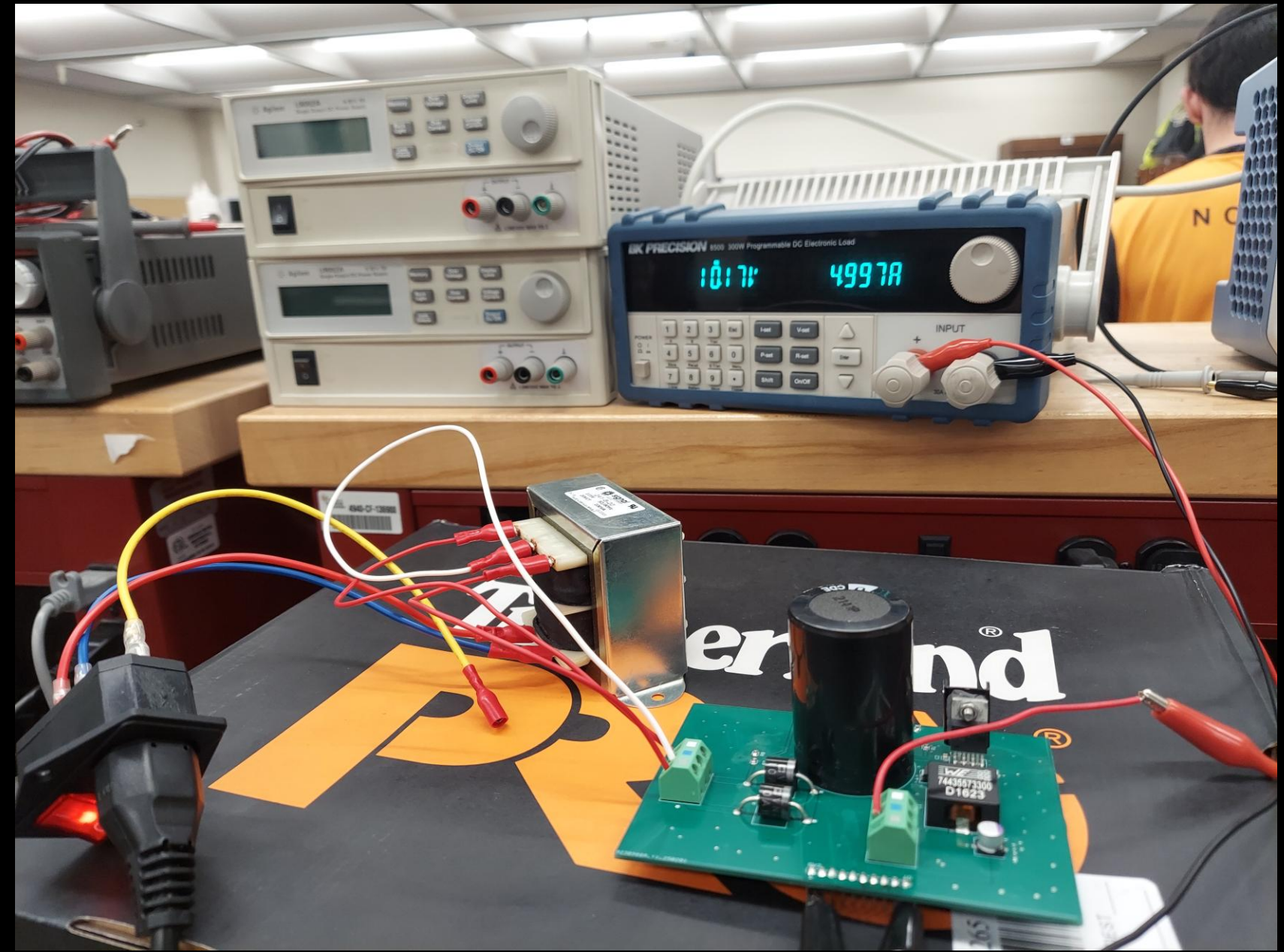
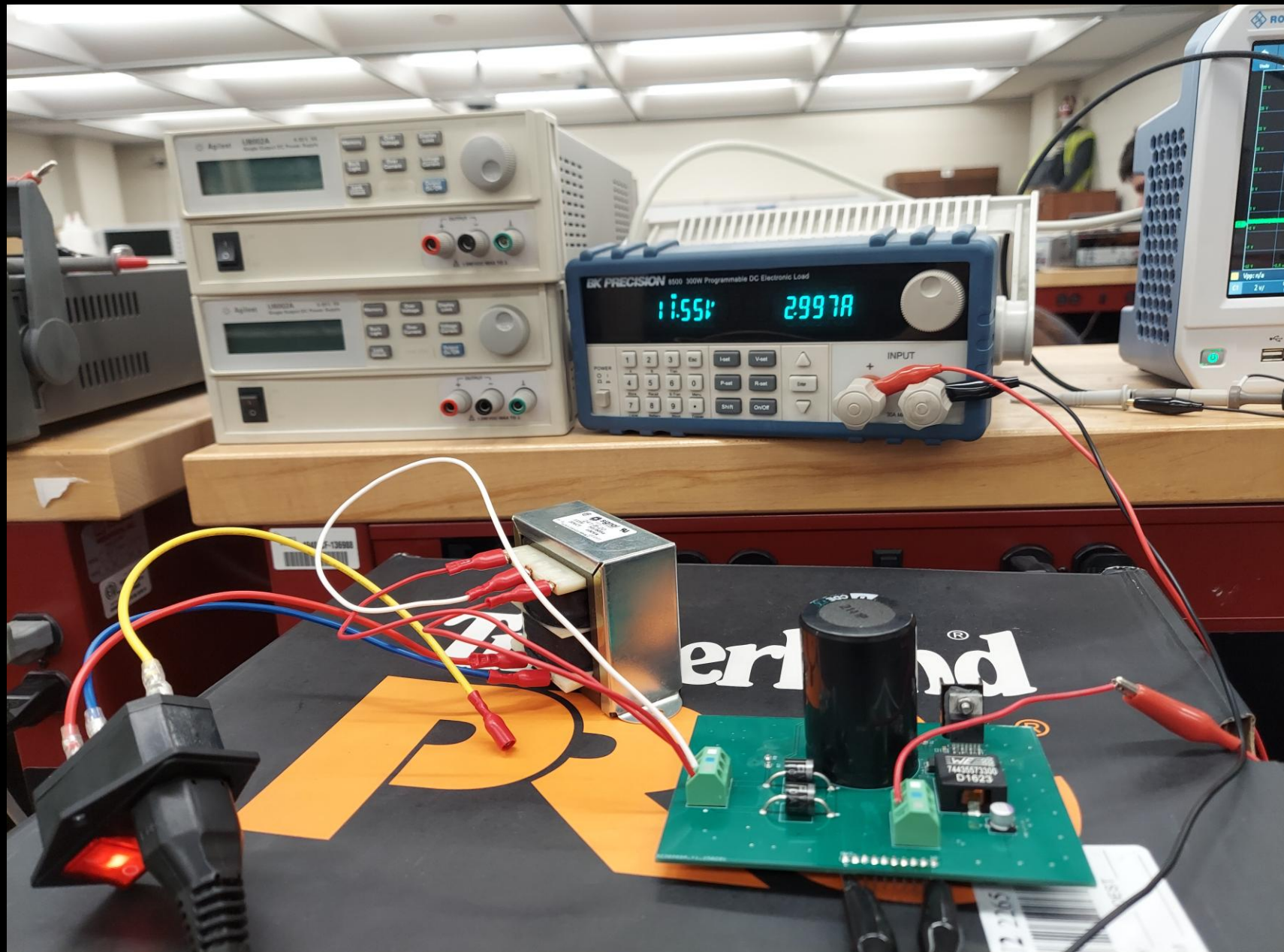


Prototyping and Testing

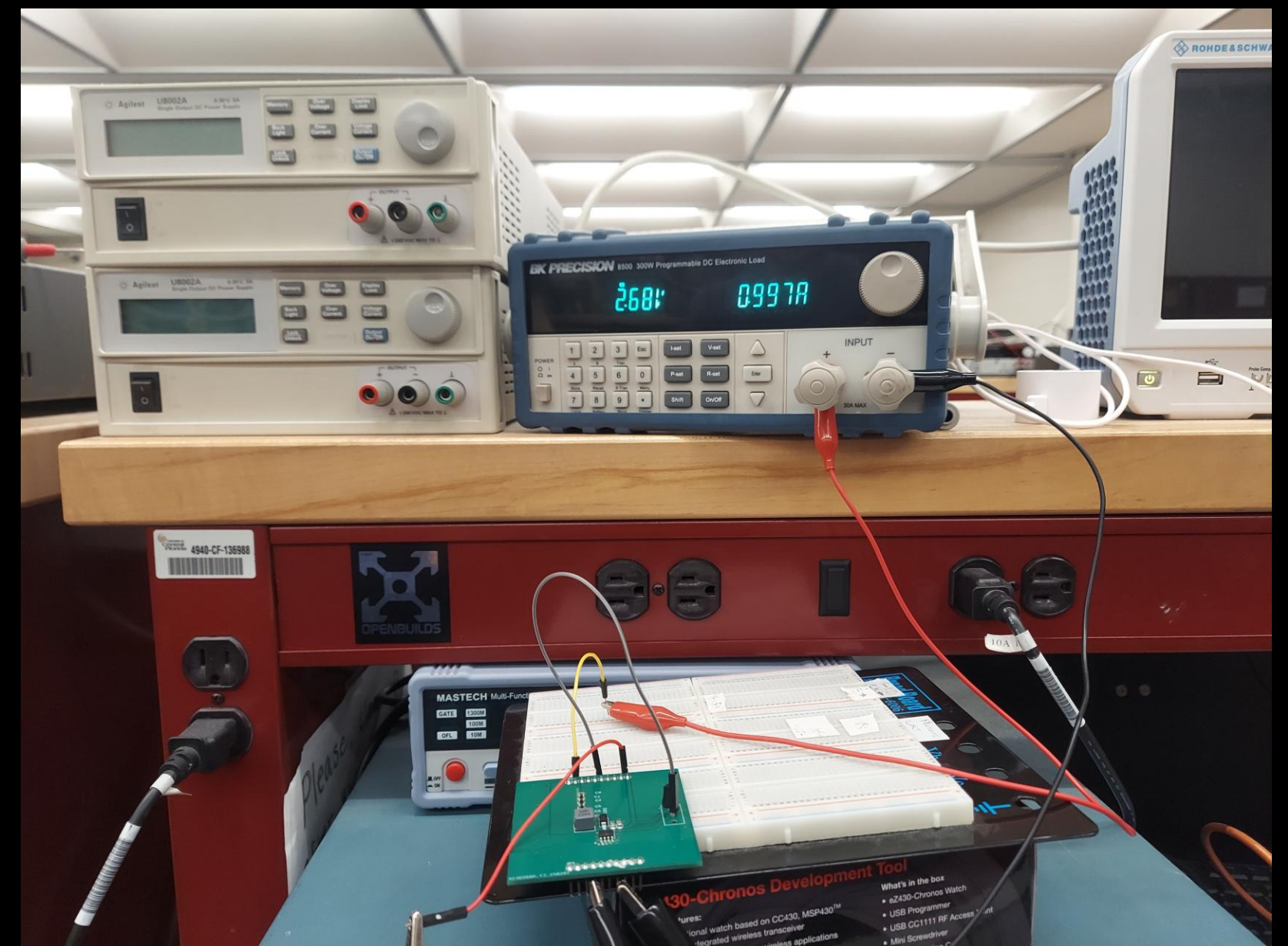
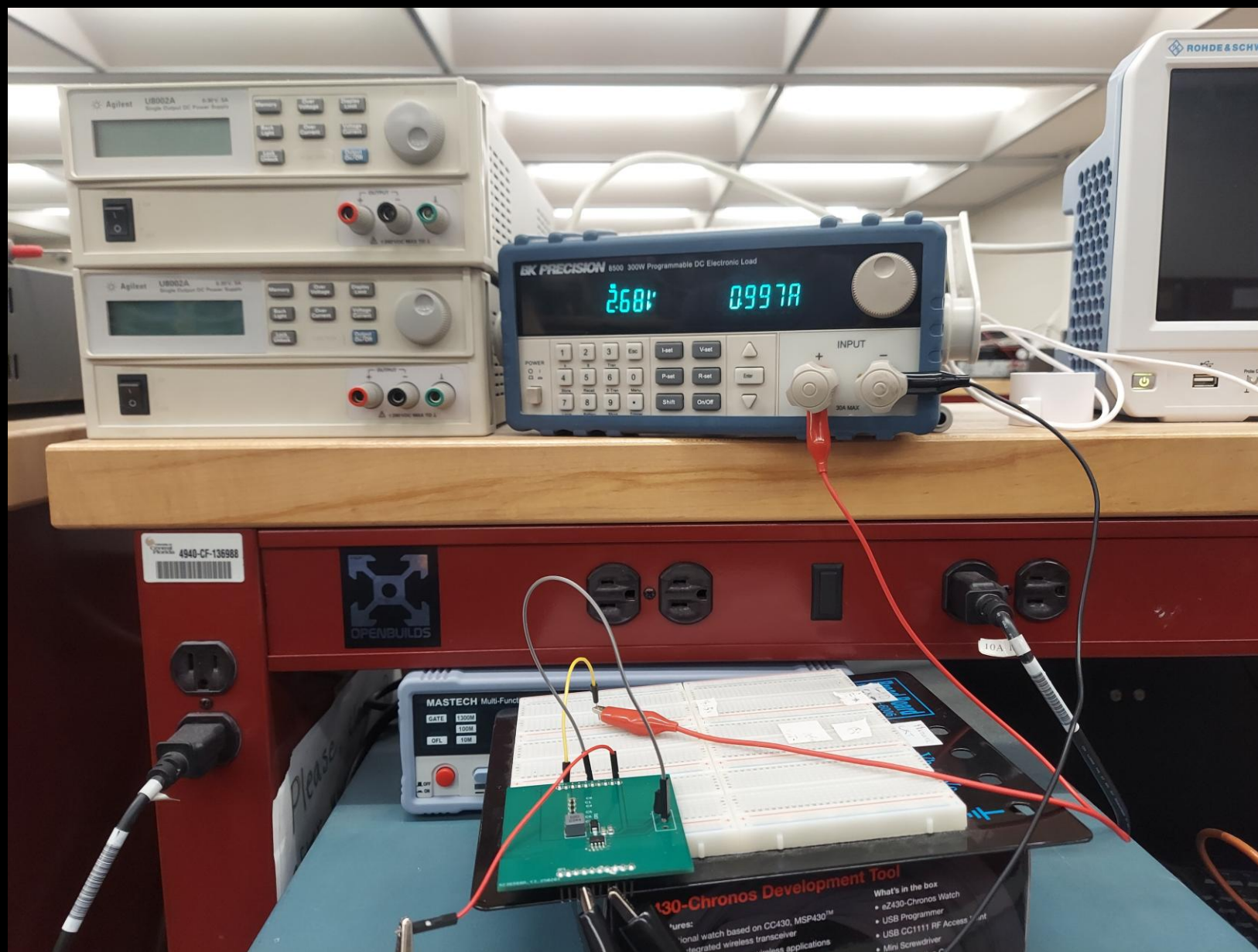
HARDWARE TESTING

- Electrical
 - 12V PSU, 3.3V PSU, and Frequency Detector are built and completely operational
 - MCU board is built and preliminarily tested, debugging WIP
 - PCBs work independently but not tested together

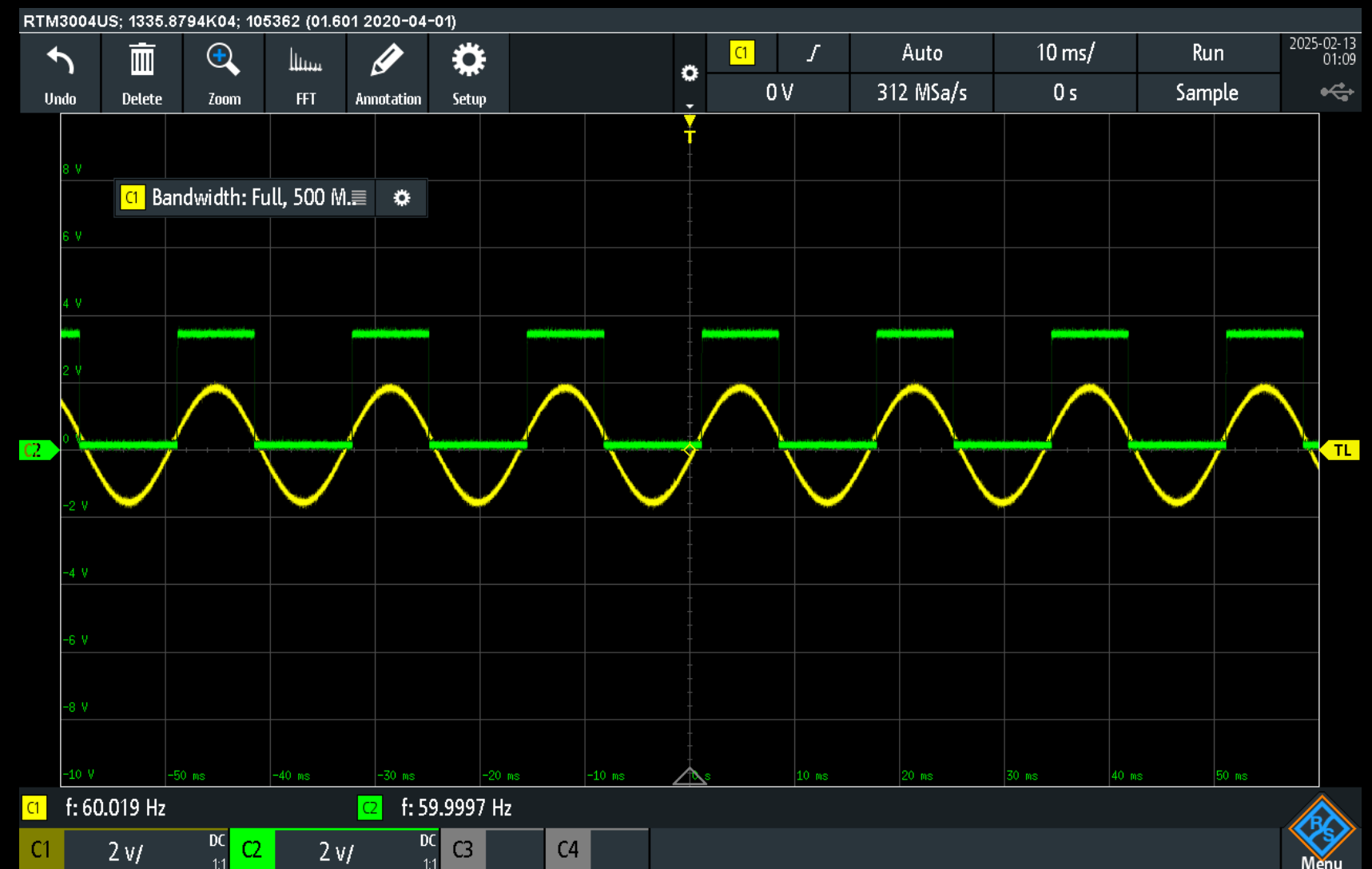
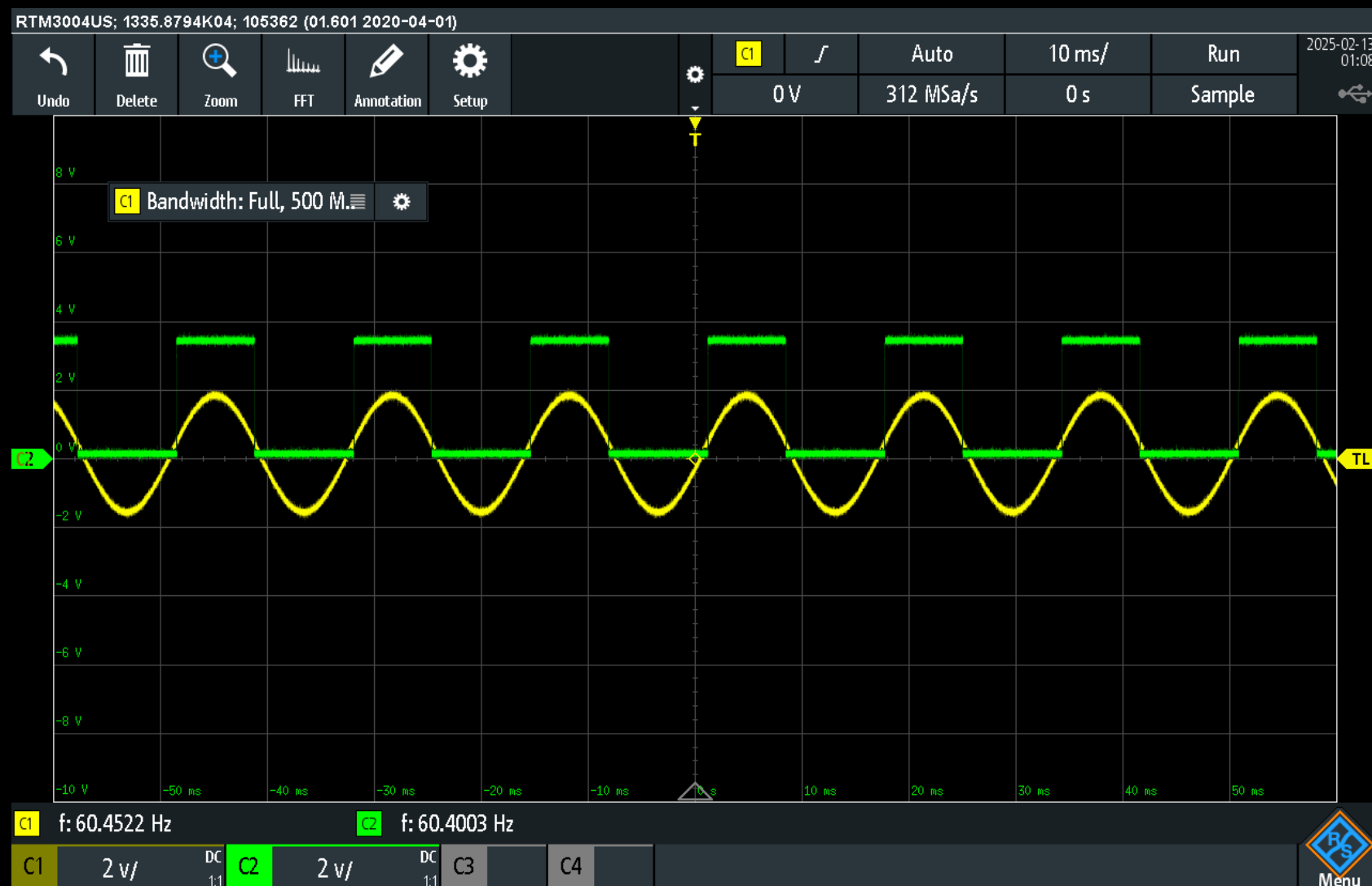
12V PSU TEST RESULTS



3.3V PSU TEST RESULTS

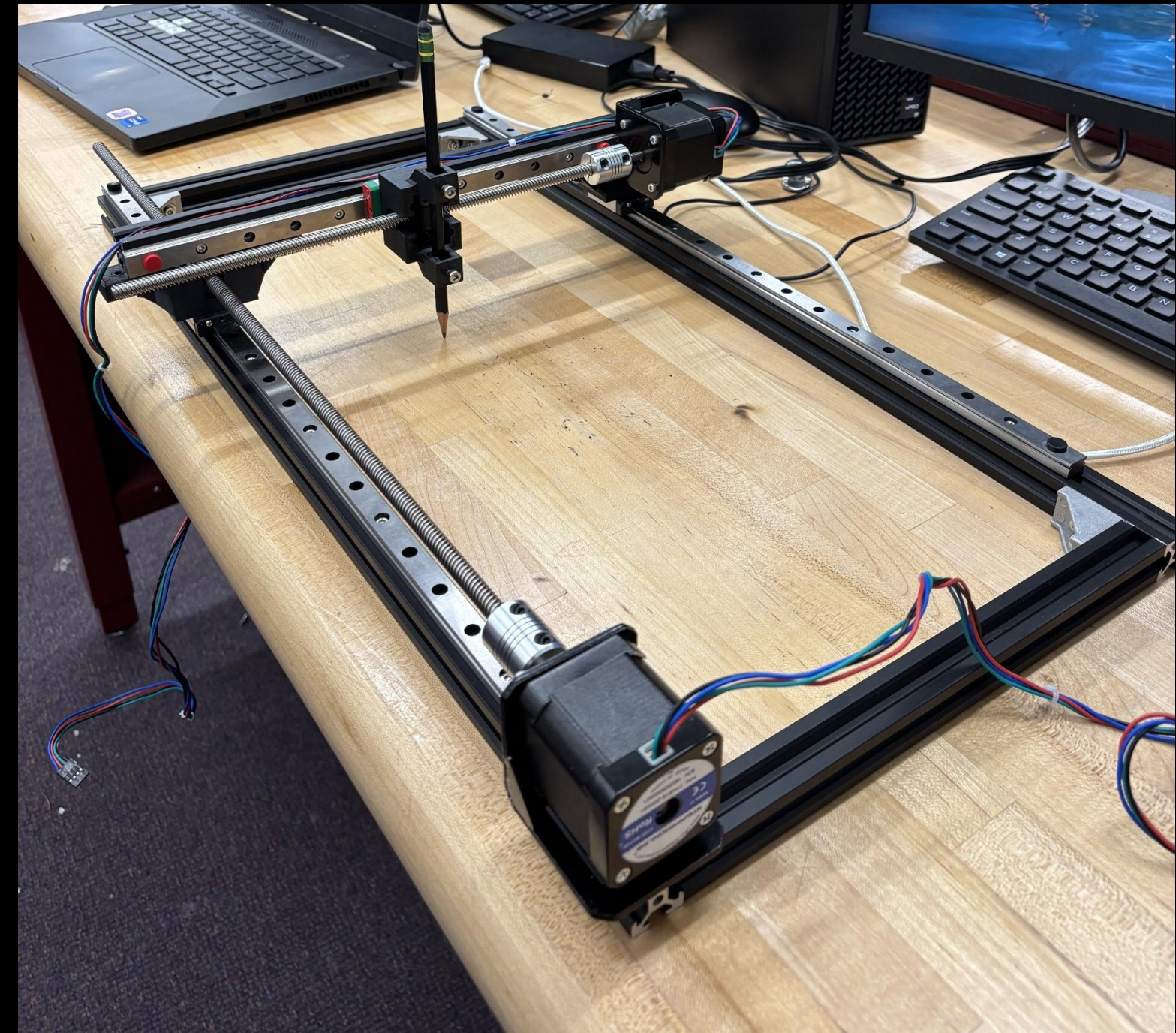


FREQUENCY DETECTOR TESTING



MECHANICAL TESTING

- 3D printed parts work, but not optimal
- Gantry runs smoothly
- End switches required for testing with power



SOFTWARE TESTING

- Software
 - App is functional and working. Needs to be polished with QOL changes upon successful API call.
 - Database is constructed and accepts insertion, deletion, indexing, and reindexing.
 - Communication tunnel is operational via both HTTP requests and Websockets.
App -> Server -> Device

Administrative Content

Budget

Sub-System	Allocated Amount
Service Panel	\$300
Electrical Hardware	\$600
Mechanical Hardware	\$400
Software	\$100
Total	\$1400

Work Distribution

Nicholas Sally

Primary Tasks:

- Frequency Sensing Design
- Power Supply Design

Secondary Tasks:

- Arm Switching Mechanism

Oscar Rodriguez

Primary Tasks:

- Mobile App Development

Secondary Tasks:

- Microcontroller Programming
- Administrative Content

Jack Dwyer

Primary Tasks:

- Gantry Switching Mechanism
- Mounting Structure

Secondary Tasks:

- Frequency Sensing Design

Abraham Alcala

Primary Tasks:

- Microcontroller Programming

Secondary Tasks:

- Mobile App Development
- Gantry Switching Mechanism

Verify and Order PCB

Done – Receive checkoff from Dr. Weeks and Submit Purchase Order

Assemble and Test PCBs

Done – Perform Continuity and Safe To Turn On (STTO) Tests

Final Checkout of Assembly

3/3/25 – Housing secured to service panel

Practice Complete Demonstration

4/7/25 – Representative Documentation of final workflow

Final Demo Day

4/12/25 – Perform and Execute a perfect demo

Timeline

Thanks For Watching!